each day. You get the picture—computers can process faster than we can, and that's why simulators are valuable.

6.5. Evolutionary algorithms as a scalable alternative

If a simulator is available, the time and financial costs of collecting samples with evolutionary algorithms is less of an issue. In fact, producing a viable agent with evolutionary algorithms can sometimes be faster than gradient-based approaches because we do not have to compute the gradients via backpropagation. Depending on the complexity of the network, this will cut down the computation time by roughly 2–3 times. But there is another advantage of evolutionary algorithms that can allow them to train faster than their gradient counterparts—evolutionary algorithms can scale incredibly well when parallelized. We will discuss this in some detail in this section.

6.5.1. Scaling evolutionary algorithms

OpenAI released a paper called "Evolutionary Strategies as a Scalable Alternative to Reinforcement Learning" by Tim Salimans et al. (2017), in which they described training agents incredibly quickly and efficiently by adding more machines. On a single machine with 18 CPU cores, they were able to make a 3D humanoid learn to walk in 11 hours. But with 80 machines (1,440 CPU cores) they were able to produce an agent in under 10 minutes.

You may be thinking that's obvious—they just threw more machines and money at the problem. But this is actually trickier than it sounds, and other gradient-based approaches struggle to scale to that many machines.

Let's first look at how their algorithm differs from what we did earlier. *Evolutionary algorithm* is an umbrella term for a wide variety of algorithms that take inspiration from biological evolution and rely on iteratively selecting slightly better solutions from a large population to optimize a solution. The approach we implemented to play CartPole is more specifically called a *genetic algorithm*, because it more closely resembles the way biological genes get "updated" from generation to generation through recombination and mutation.